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Excerpt
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THE THIRTIETH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

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A-1. Let $f(x, y)$ be a polynomial with real coefficients in the real variables x and y defined over the entire x-y plane. What are the possibilities for the range of $f(x, y)$?

A-2. Let D_n be the determinant of order n of which the element in the ith row and the jth column is the absolute value of the difference of i and j. Show that D_n is equal to

$$(-1)^{n-1}(n-1)2^{n-2}.$$

A-3. Let P be a non-self-intersecting closed polygon with n sides. Let its vertices be $P_1, P_2, \cdots, P_n$. Let m other points, $Q_1, Q_2, \cdots, Q_m$ interior to P be given. Let the figure be triangulated. This means that certain pairs of the $(n+m)$ points $P_1, \cdots, Q_m$ are connected by line segments such that (i) the resulting figure consists exclusively of a set T of triangles, (ii) if two different triangles in T have more than a vertex in common then they have exactly a side in common, and (iii) the set of vertices of the triangles in T is precisely the set of $(n+m)$ points $P_1, \cdots, Q_m$. How many triangles in T?

A-4. Show that

$$\int_0^1 x^x dx = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-n}.$$

(The integrand is taken to be 1 at $x=0$.)

A-5. Let $u(t)$ be a continuous function in the system of differential equations

$$\frac{dx}{dt} = -2y + u(t), \qquad \frac{dy}{dt} = -2x + u(t).$$

Show that, regardless of the choice of $u(t)$, the solution of the system which satisfies $x=x_0$, $y=y_0$ at $t=0$ will never pass through (0, 0) unless $x_0=y_0$. When $x_0=y_0$, show that, for any positive value t_0 of t, it is possible to choose $u(t)$ so the solution is at (0, 0) when $t=t_0$.

A-6. Let a sequence $\{x_n\}$ be given, and let $y_n=x_{n-1}+2x_n$, $n=2, 3, 4, \cdots$. Suppose that the sequence $\{y_n\}$ converges. Prove that the sequence $\{x_n\}$ also converges.

B-1. Let n be a positive integer such that $n+1$ is divisible by 24. Prove that the sum of all the divisors of n is divisible by 24.

B-2. Show that a finite group can not be the union of two of its proper subgroups. Does the statement remain true if "two" is replaced by "three"?

B-3. The terms of a sequence T_n satisfy

$$T_nT_{n+1} = n \qquad (n = 1, 2, 3, \cdots) \text{ and } \lim_{n\to\infty} \frac{T_n}{T_{n+1}} = 1.$$

Show that $\pi T_1^2=2$.

B-4. Show that any curve of unit length can be covered by a closed rectangle of area 1/4.